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Department of
Artificial Intelligence & Machine Learning

AML23702
Advance Natural Language Processing

AML23702_Syllabus.pdf

VASUGI I

ASSISTANT PROFESSOR

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Module-2

Semantic Web Ontology and Knowledge Graph

Semantic web ontology and Knowledge Graph: Introduction to semantic web, Semantic web ontology, Semantic web languages, Ontology Engineering, Ontology Learning, Knowledge graph –construction of graph

Reference : *Thinking_on_the_Web.pdf*

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Pre-Session Reading

- An introduction to the semantic web, why it was proposed, and how it helps machines understand web data through explicit meaning and structure - https://milvus.io/ai-quick-reference/what-is-the-purpose-of-semantic-web-in-the-context-of-knowledge-graphs
- A resource on the steps of building a knowledge graph from domain modeling, extraction, linking, and storage - https://neo4j.com/blog/knowledge-graph/how-to-build-knowledge-graph/
- A domain-focused article showing how semantic web and KGs support enterprise search, integration, and reasoning- https://www.dataversity.net/articles/semantic-web-and-ai-empowering-knowledge-graphs-for-smarter-applications/

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Semantic Web & Knowledge Graphs

From linked documents to machine-processable meaning

LEARNING OUTCOMES

1. Explain RDF graphs and the Semantic Web stack
2. Distinguish taxonomy, ontology, and knowledge graph
3. Select languages for modeling, querying, and validation

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The Web needs meaning, not only links

WEB OF DOCUMENTS

Designed for people to read

- Links connect pages
- Meaning is implicit in text and layout
- Integration depends on custom parsing

→

WEB OF DATA

Built for people + machines

- IRIs (Internationalized Resource Identifier) identify things globally
- Typed links express relationships
- Shared vocabularies make meaning portable

Semantic Web = linked data + explicit semantics + interoperable standards

Source: W3C Semantic Web Standards

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The Semantic Web is a layered standards stack

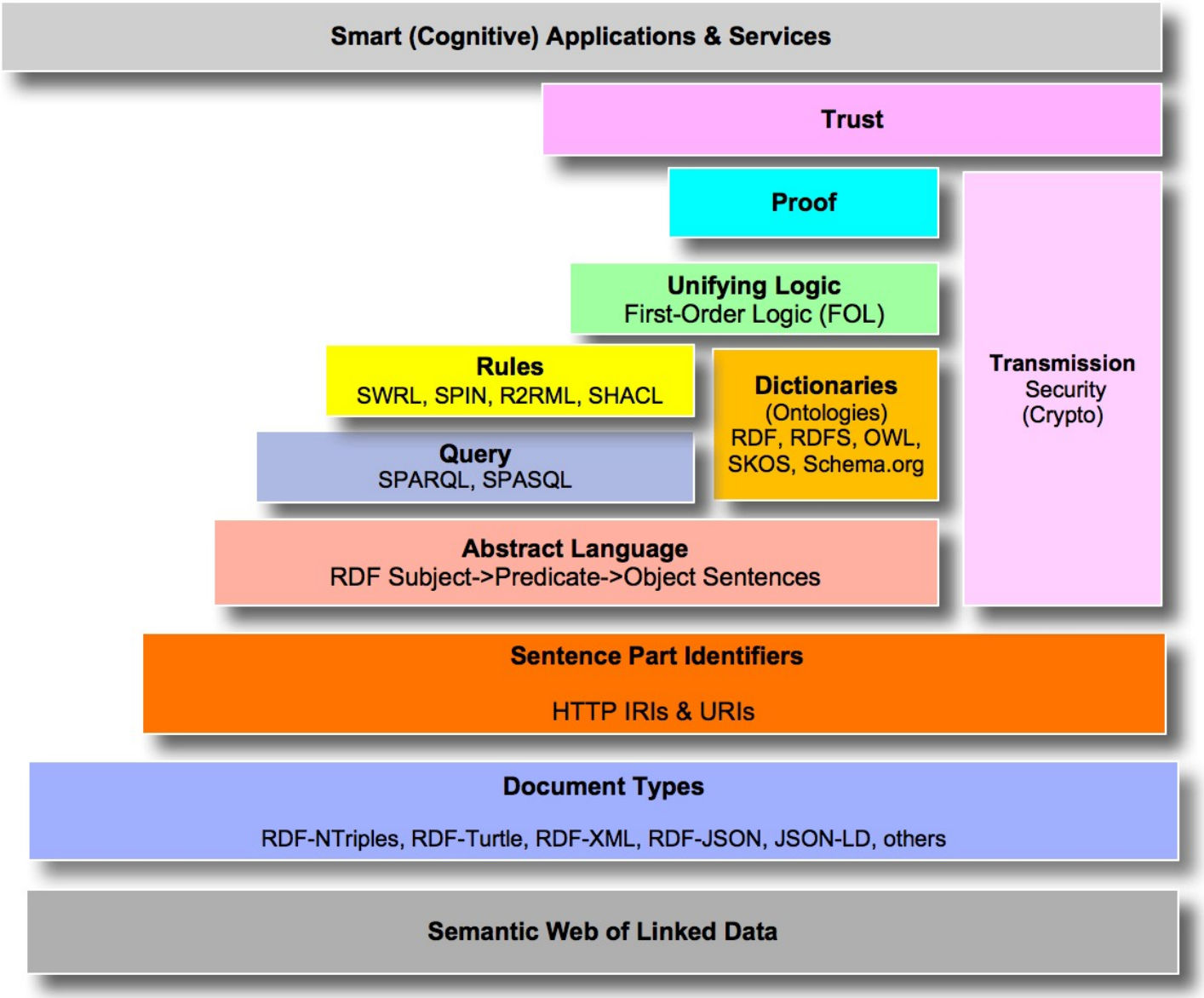
TRUST, PROVENANCE & APPLICATION RULES	Who said it? Can it be trusted?
SHACL • RULES • REASONING	Validate constraints and derive consequences
SPARQL	Query and update graph content
RDFS • OWL • SKOS	Define vocabularies, constraints, and conceptual meaning
RDF GRAPH + SERIALIZATIONS	Represent facts as subject-predicate-object triples
IRIs • LITERALS • DATATYPES	Name things and express values consistently

Higher layers depend on lower layers; deployments choose only what they need.

Sources: W3C RDF 1.1; RDFS 1.1; OWL 2; SPARQL 1.1; SHACL

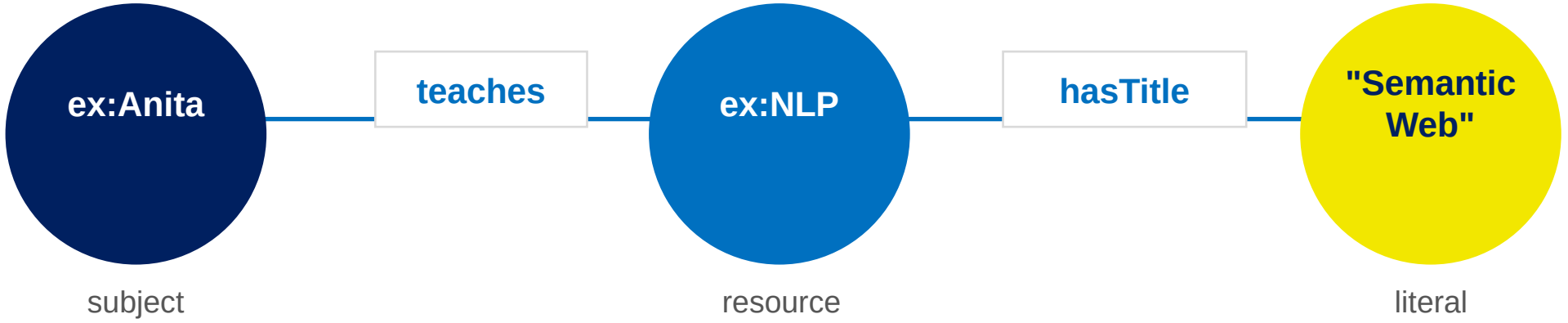
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The Semantic Web layered architecture



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RDF expresses knowledge as a labeled graph



One triple

```
ex:Anita   ex:teaches   ex:NLP .
```

RDF terms

IRI • blank node • literal
Named graphs organize triples into datasets

Source: W3C RDF 1.1 Concepts and Abstract Syntax

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Ontologies make a domain's meaning explicit

CLASSES
Faculty • Course

PROPERTIES
teaches • hasTitle

INDIVIDUALS
Anita • NLP

AXIOMS
Domain • range • disjoint

Ontology = a shared vocabulary + formal statements about how domain concepts relate

Example axioms

```
teaches  domain  Faculty
teaches  range   Course
Faculty  disjointWith  Course
```

Result: software can check consistency and infer facts not stated explicitly.

Sources: Stanford Ontology Development 101; W3C OWL 2 Overview

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Taxonomy ≠ ontology ≠ knowledge graph

TAXONOMY

Organizes terms

```
Course
↳ AI Course
   ↳ AML23702
```

Hierarchy

ONTOLOGY

Defines meaning

```
Course ⊑ LearningUnit
Faculty ⊑ Person
Faculty —teaches→ Course
```

Schema + constraints

KNOWLEDGE GRAPH

Populates connected facts

```
Anita —teaches→ AML23702
AML23702 —hasTitle→"Semantic Web"
```

Entities + relationships + context

Progression: organize → formalize → connect real-world instances

Sources: Stanford Ontology Development 101; Hogan et al., Knowledge Graphs (2021)

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RDFS and OWL add increasing semantic expressivity

RDF
Facts as triples
type • property • value

RDFS
Vocabulary structure
class • subclass • domain • range

OWL 2
Formal ontology
equivalence • disjointness • restrictions

More expressivity → richer inference → higher reasoning cost

OWL assumptions to remember

Open world: missing facts are unknown, not false.

No unique-name assumption: two IRIs may denote the same thing.

Sources: W3C RDF Schema 1.1; OWL 2 Overview and Primer

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Semantic Web languages form a toolchain

DESCRIBE	MODEL	QUERY	VALIDATE
RDF	**RDFS / OWL**	**SPARQL**	**SHACL**
Graph data model	**Vocabulary & ontology**	**Retrieve & update**	**Check graph quality**
Turtle • JSON-LD RDF/XML • RDFa	Classes • axioms concept schemes	Patterns • filters aggregation • federation	Shapes • constraints validation reports

Publish & exchange HTTP IRIs • content negotiation • named graphs • provenance (PROV)

Serialization is syntax; RDF is the underlying graph model.

Sources: W3C Semantic Web Standards; RDF 1.1; OWL 2; SPARQL 1.1; SHACL

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RDF states facts; RDFS and OWL add meaning

RDF

State facts

```
ex:Anita
  ex:teaches
  ex:AML23702 .

ex:AML23702
  ex:hasTitle
  "Semantic
Web" .
```

Two explicit triples

RDFS

Define vocabulary

```
ex:teaches
  rdfs:domain ex:Faculty
;
  rdfs:range
ex:Course .

  ex:Faculty
    rdfs:subClassOf
    ex:Person .
```

Infers types and hierarchy

OWL 2

Add formal meaning

```
ex:teaches
  owl:inverseOf
  ex:isTaughtBy .

ex:Faculty
  owl:disjointWith
  ex:Course .
```

Infers inverse links; checks conflicts

Result: Anita is inferred as Faculty; AML23702 as Course; and AML23702 isTaughtBy Anita.

Sources: W3C RDF 1.1; RDF Schema 1.1; OWL 2 Primer

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Linked Data makes knowledge graphs discoverable

1 Use IRIs as names
Identify any thing or concept

→ **2 Use HTTP IRIs**
Make identifiers look-up-able

→ **3 Return useful data**
Provide RDF and query access

→ **4 Link to other IRIs**
Enable discovery across datasets

Publishing pipeline

Name → **Describe** → **Serve** → **Link** → **Discover**

The value grows when independently published graphs share identifiers and links.

Source: Tim Berners-Lee, Linked Data - Design Issues (2006)

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Building a knowledge graph is an iterative lifecycle



Start with questions, not technology. A useful ontology is scoped by the queries, decisions, and validations the graph must support.

Sources: Stanford Ontology Development 101; Hogan et al., Knowledge Graphs (2021)

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Knowledge graphs improve through iteration

FROM QUESTION TO GRAPH

1	**Ask**	Define competency questions, users, decisions and success criteria.
2	**Model**	Define classes, properties, axioms, constraints and an IRI policy.
3	**Ingest**	Normalize evidence; preserve source, timestamp and confidence.
4	**Link**	Resolve duplicates; assign canonical IRIs and retain source keys.
5	**Validate**	Run reasoning, SHACL and quality checks to expose conflicts and gaps.
6	**Use**	Run queries and applications; record failures and unanswered questions.
7	**Refine**	Revise questions, schema, mappings, identity and thresholds; version changes.

WHY THE LOOP MUST REPEAT

- **Requirements evolve:** New users and questions expose missing concepts, relations or constraints.
- **Sources change:** New datasets and schema drift break mappings, identifiers and datatype assumptions.
- **Errors become evidence:** False merges, contradictions and SHACL violations reveal defects upstream.
- **Use reveals fitness:** Real queries expose coverage, usability and performance gaps that design alone misses.

Feedback closes the loop. Evidence from Validate and Use returns to Ask and Model because requirements, data and domain knowledge evolve—and early assumptions are rarely complete.

Sources: Stanford Ontology Development 101; Hogan et al., Knowledge Graphs (2021); W3C SHACL

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Query, infer, and validate the same graph

1 ASSERTED DATA

```
ex:Anita  ex:teaches  ex:AML23702 .
ex:AML23702  ex:hasTitle  "Semantic
Web" .
```

2 ONTOLOGY

```
ex:teaches  rdfs:domain
ex:Faculty .
ex:teaches  rdfs:range   ex:Course .
```

3 INFERRED

XYZ is a Faculty • AML23702 is a Course

SPARQL QUERY

```
SELECT ?course ?title
WHERE {
  ex:Anita ex:teaches ?course .
  ?course ex:hasTitle ?title .
}
```

RESULT

AML23702	**|**	**Semantic Web**

SHACL CHECK Every Course must have exactly one title.

Sources: W3C SPARQL 1.1; OWL 2; SHACL

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Semantic Web value must exceed modeling cost

WHERE IT FITS

Data integration — Unify heterogeneous sources through shared identifiers

Semantic search & QA — Retrieve by entities and relationships, not keywords alone

AI grounding — Give agents and RAG systems structured, traceable context

Compliance & lineage — Represent provenance, policy, and dependencies

DESIGN TRADE-OFFS

- Ontology design requires domain expertise
- Identity resolution and data quality remain hard
- Expressive reasoning can be computationally expensive
- Governance is needed as vocabularies evolve

Name things → connect facts → define meaning → query, infer, and validate

Sources: Hogan et al., Knowledge Graphs (2021); W3C Semantic Web Standards

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OWL properties control links, values, and inference

OWL construct	Formal meaning	Example	Reasoning effect
Object property	Links one individual to another	Anita teaches Course1	Models entity relationships
Datatype property	Links an individual to a literal	Course1 credits “3” xsd:integer	Models typed attributes
Annotation property	Adds labels, comments, or provenance	Course rdfs:label “Course”	Descriptive; normally non-logical
Functional property	At most one value for each subject	hasBirthDate	Multiple values must agree
Inverse-functional	At most one subject for each object	hasEmployeeRecord	Shared value can imply same subject
Transitive property	$P(a,b) \land P(b,c) \Rightarrow P(a,c)$	partOf	Derives links across a chain
Symmetric property	$P(a,b) \Rightarrow P(b,a)$	collaboratesWith	Adds the reverse relation
Asymmetric property	$P(a,b) \Rightarrow$ not $P(b,a)$	supervises	Reverse assertion is inconsistent
owl:inverseOf	$P(a,b) \Leftrightarrow Q(b,a)$	teaches / isTaughtBy	Derives a named inverse relation

Sources: W3C OWL 2 Primer; OWL 2 Structural Specification

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Ontology Engineering & Ontology Learning

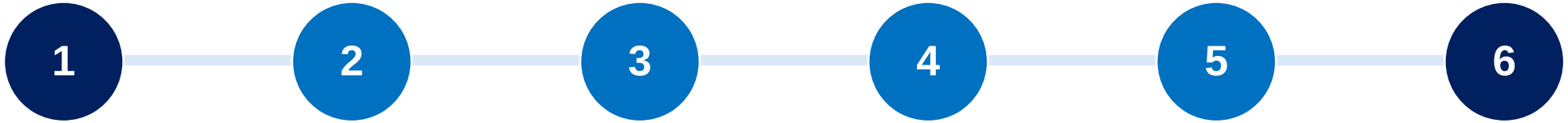
Design disciplined models — accelerate discovery from evidence

LEARNING OUTCOMES

1. Plan an ontology engineering lifecycle
2. Evaluate models for correctness and usefulness
3. Explain ontology learning tasks, methods, and human oversight

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Ontology engineering creates an evolving formal asset



Specify	Model	Formalize	Implement	Evaluate	Maintain
Purpose, users, scope	Terms, relations, constraints	OWL axioms & patterns	Encode, document, version	Reason, test, review	Change, align, govern

Iterative, not waterfall. Evaluation and stakeholder feedback can reopen scope, modeling, and implementation decisions.

Sources: Noy & McGuinness (2001); Gómez-Pérez et al. (2004)

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Competency questions keep ontology scope testable

REQUIREMENTS SPECIFICATION

Purpose	Why must the ontology exist?
Users	Who will create, query, or maintain it?
Scope	Which concepts and boundaries belong?
Tests	Which questions and constraints must it support?

COMPETENCY QUESTIONS

Which faculty member teaches AML23702?

Which courses cover Semantic Web topics?

Can one course belong to multiple modules?

Each question becomes a query, example, or validation test.

Sources: Noy & McGuinness (2001); Suárez-Figueroa et al. (2009)

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Model for clarity, coherence, and change

CLARITY	Definitions communicate intended meaning
COHERENCE	Axioms do not contradict one another
EXTENDIBILITY	New concepts fit without redesign
MINIMAL COMMITMENT	Assert only what the task requires

REUSE BEFORE REINVENTION

Adopt established vocabularies; import or align only the terms needed.

COMMON ANTI-PATTERNS

- Class vs. instance confusion
- Overusing owl:sameAs
- Deep hierarchies without a use case

Sources: Gruber (1995); Noy & McGuinness (2001); W3C OWL 2

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Evaluate both correctness and fitness for use

LOGICAL	STRUCTURAL	DOMAIN	TASK
Consistency	**Model quality**	**Coverage**	**Fitness for use**
Can a reasoner find contradictions?	Are hierarchy and relations coherent?	Does it represent expert knowledge?	Does it answer required questions?
Unsatisfiable classes	Cycles, redundancy	Missing or wrong concepts	Query and SHACL tests

Release gate: no logical blockers, agreed domain coverage, passing competency questions, and documented limitations.

Sources: Brank et al. (2005); Gómez-Pérez et al. (2004)

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Ontology learning proposes structure from evidence

INPUT EVIDENCE

Text documents • definitions • manuals

Data tables • schemas • logs

Prior models taxonomies • graphs • lexicons

→

LEARNED CANDIDATES

Terms “course”, “module”, “faculty”

Concepts Course • Module • Faculty

Taxonomy `Course ⊑ LearningUnit`

Relations `Faculty teaches Course`

Axioms `domain • range • disjointness`

Learning accelerates discovery; it does not automatically establish truth.

Sources: Maedche (2002); Cimiano (2006); Shamsfard & Barforoush (2003)

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Ontology learning methods have distinct trade-offs

LINGUISTIC PATTERNS	STATISTICAL	EMBEDDINGS	LLM-ASSISTED
Hearst-style and dependency patterns	Frequency, association, clustering	Semantic similarity and graph vectors	Prompted extraction and candidate generation
STRENGTH	**STRENGTH**	**STRENGTH**	**STRENGTH**
Interpretable; precise	Scales to large corpora	Finds latent relationships	Flexible and fast to prototype
LIMIT	**LIMIT**	**LIMIT**	**LIMIT**
Limited recall	Needs thresholds and clean data	Harder to explain	May hallucinate; requires evidence

Best practice: combine methods and rank candidates by confidence + supporting evidence.

Sources: Maedche (2002); Cimiano (2006); Shamsfard & Barforoush (2003)

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Human review governs learned ontology candidates

1	2	3	4	5	6
Extract	**Propose**	**Review**	**Formalize**	**Test**	**Publish**
“faculty teaches course”	teaches(Faculty, Course)	expert confirms meaning	domain/range in OWL	queries + consistency	version + provenance

MACHINE Find patterns, rank candidates, surface evidence

HUMAN Resolve ambiguity, approve meaning, own governance

Continuous loop: engineering defines quality; learning supplies candidates; use and feedback drive the next revision.

Sources: Noy & McGuinness (2001); Cimiano (2006); W3C OWL 2

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Worked example | University course ontology

1 · DEFINE THE MODEL

owl:Thing

- **Person**
  - Student · Faculty
- **Course**
- **Department**

Object properties

enrolledIn Student → Course

teaches Faculty → Course

offeredBy Course → Dept.

2 · POPULATE WITH INDIVIDUAL FACTS

SUBJECT	PROPERTY	OBJECT
John Student	— enrolledIn →	**ArtificialIntelligence** Course
Dr_Ravi Faculty	— teaches →	**ArtificialIntelligence** Course
ArtificialIntelligence Course	— offeredBy →	**ComputerScience** Department

3 · ADD OWL SEMANTICS + REASON

Student SubClassOf enrolledIn min 1 Course

The reasoner can check the constraint and infer additional class membership.

Why it is an ontology: formal relations + constraints support machine inference—not just classification.

Worked example adapted from the University Course Ontology material on this slide.

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Knowledge Graph Construction

From heterogeneous evidence to a governed, queryable graph

LEARNING OUTCOMES

1. Explain the end-to-end construction pipeline
2. Create mapping, identity, and validation rules
3. Plan publication, provenance, and graph maintenance

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A knowledge graph is built as a controlled pipeline



1	2	3	4	5	6	7
Define	**Acquire**	**Extract**	**Resolve**	**Transform**	**Validate**	**Publish**
use cases & contract	source evidence	entities & relations	identity & links	mapping to triples	quality & constraints	serve & observe

Construction is iterative. New sources, errors, and user feedback send the pipeline back to the contract, schema, mappings, or identity rules.

Sources: Hogan et al. (2021); W3C RDF 1.1; SHACL

Knowledge Graph Construction Pipeline

(Example: Healthcare)

1 Define use cases and requirements

Example questions:

- Which doctors treat which conditions?
- Which patients are at high risk for drug interactions?

Identify key entities and relationships:

- Patient
- Doctor
- Condition
- Drug

Define constraints (e.g., valid links).

2 Acquire relevant source evidence

- EHR database
- Clinical guidelines (PDF/HTML)
- Staff directory (CSV)
- Clinical trial reports (text)

Note:
record formats, update frequency, access controls.

3 Extract entities and relationships

Example text:

"Dr. Alice Kumar, a cardiologist at City Hospital, prescribed Lisinopril to patient John Doe for hypertension."

Extracted entities:

- Doctor: Alice Kumar
- Patient: John Doe
- Drug: Lisinopril
- Condition: Hypertension
- Organization: City Hospital

Extracted relationships:

- worksAt
- treats
- prescribedDrug
- hasCondition
- treatsCondition

4 Resolve duplicate identities (entity resolution)

Record A
John Doe
DOB: 1980-05-12
MRN: 123456

Record B
J. Doe
DOB: 1980-05-12
Phone: 555-1234

Canonical entity
Patient: JohnDoe_123456
Aliases: "John Doe", "J. Doe"
(merged attributes)

5 Transform data into triples

Example triples (RDF):

```
:JohnDoe_123456  a
:Patient .

:JohnDoe_123456
:hasCondition
:Hypertension .

:AliceKumar_789  a
:Doctor .

:AliceKumar_789
:worksAt
:CityHospital .

:AliceKumar_789
:prescribedDrug
:Lisinopril .

:Lisinopril  a  :Drug .

:Lisinopril
 treatsCondition
:Hypertension .
```

6 Validate quality and constraints

Example checks:

- prescribedDrug links a Patient to a Drug
- Doctor has at least one worksAt relationship
- hasCondition links to a known Condition
- No duplicate patient IDs
- Use controlled vocabularies

Issues found (example):

- Misspelled drug name
- Doctor with no hospital

Fix mappings / rules and re-run.

7 Publish and monitor the graph

- Load into graph database (e.g., Neo4j)
- Expose APIs (SPARQL / Cypher)
- Use in applications (search, decision support, analytics)
- Monitor:
  - new/updated triples
  - validation errors
  - user feedback

Continuous improvement
(update extraction, resolution rules, ontology, etc.)

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Define the graph contract before ingesting data

PURPOSE	Questions and decisions the graph must support	**SCOPE**	Included domains, time ranges, and source boundaries
IDENTITY	IRI patterns, keys, and duplicate policy	**SCHEMA**	Classes, properties, datatypes, and constraints
PROVENANCE	Source, extraction activity, version, and confidence	**QUALITY**	Acceptance thresholds and ownership

A contract prevents “load first, model later” from becoming permanent technical debt.

Sources: Noy & McGuinness (2001); Hogan et al. (2021); W3C PROV-O


1. Purpose
What decisions and questions must the graph support?
Example: Which faculty teaches which course? Which students satisfy prerequisities?
2. Scope
What is in and what is out?
Example: Courses, faculty, students, departments (2020–2026)
Out of scope: research projects, alumni donations
3. Identity
Rules for IRIs, keys, and duplicate entities
Example: student_id, employee_id
Merge duplicates (e.g., same student with name variations)
Flag conflicts for review
4. Schema
Classes, properties, datatypes, and constraints
Example: Student, Course, FacultyMember, Department
Relationships: teaches, offeredBy, hasPrerequisite, enrolledIn
Constraints: valid links, no cycles
5. Provenance
How sources, versions, and confidence are recorded
Example: source system, timestamp, confidence score
Trace errors and support audits
6. Quality
Acceptance thresholds and ownership
Example: ≥ 95% courses have department, data updated within 24 hours
Owners: Registrar, HR, SIS team
Validation: automated tests, schema constraints
Knowledge Graph Contract
(Design & Governance Agreement)
Before collecting data, we define what the graph must do and how it will be governed.

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Normalize sources into a common evidence layer

SOURCE SYSTEMS

Relational	`course_id • faculty_id`
Documents	`syllabus • handbook`
APIs / JSON	`catalog • people`
Existing graphs	`RDF • SKOS • OWL`

→

COMMON EVIDENCE LAYER

source_id	where the record came from
record_id	stable source key
raw_value	original evidence
normalized_value	cleaned representation
timestamp	validity / extraction time
confidence	optional extraction score

Keep raw evidence alongside normalized values so every graph assertion remains traceable.

Sources: Hogan et al. (2021); W3C R2RML; JSON-LD 1.1

Common Evidence Layer in Knowledge Graph Construction

(Example: Faculty Name)

1. HR System (Relational Table)

employee_id	name	department
12345	Dr. Anita Rao	Computer Sci

source_id: HR_DB
record_id: EMP_12345
raw_value: "Dr. Anita Rao"

2. Course Catalog (JSON)

```
{
  "instructor_id": "INS_77",
  "instructor_name": "Anita Rao",
  "courses": ["CS101", "CS202"]
}
```

source_id: COURSE_CATALOG_JSON
record_id: INS_77
raw_value: "Anita Rao"

3. Existing Graph (RDF)

```
<http://old-graph.example.org/
 faculty/Anita_Rao>
 foaf:name "Anita_Rao" .
```

source_id: LEGACY_RDF
record_id: <.../faculty/Anita_Rao>
raw_value: "Anita_Rao"

Common Evidence Layer

Stores all extracted values with source details (before merging into the graph)

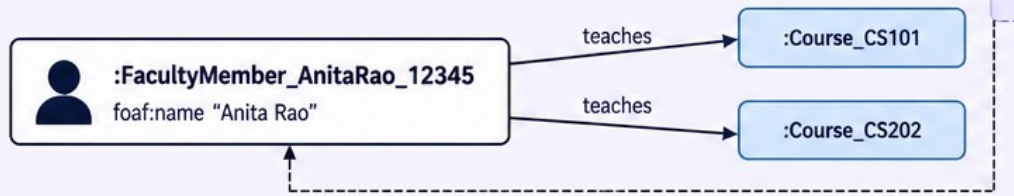
source_id	record_id	raw_value	normalized_value	timestamp	confidence
HR_DB	EMP_12345	Dr. Anita Rao	Anita Rao	2025-09-01T08:30Z	0.98
COURSE_CATALOG_JSON	INS_77	Anita Rao	Anita Rao	2025-09-01T09:15Z	0.95
LEGACY_RDF	<.../faculty/Anita_Rao>	Anita_Rao	Anita Rao	2025-09-01T10:00Z	0.90

What does the evidence layer do?

- Keeps the original values from all sources
- Stores normalized values
- Records source, timestamp and confidence
- Helps with debugging, provenance and user explanation

Final Knowledge Graph

Clean, normalized entities and relationships



But we can explain where the data came from:

Name appears as:

- "Dr. Anita Rao" in HR_DB (EMP_12345)
- "Anita Rao" in COURSE_CATALOG_JSON (INS_77)
- "Anita_Rao" in LEGACY_RDF (<.../faculty/Anita_Rao>)

The evidence layer retains the raw evidence; the graph provides a clean, unified view.

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Extract candidate entities and relations from evidence

STRUCTURED PATH

Select rows → **Map columns** → **Create candidates**

filter + join | keys + datatypes | entities + edges

UNSTRUCTURED PATH

Detect mentions → **Classify relations** → **Attach evidence**

NER / terms | patterns / models | span + score

CANDIDATE ASSERTION

ex:Anita ex:teaches ex:AML23702

evidence: faculty_assignment row 42
• confidence: 0.98

Candidates are not final facts until identity, schema, and quality checks pass.

Sources: Hogan et al. (2021); Cimiano (2006)

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Resolve identity before connecting data across sources

SOURCE A

faculty:17
Dr. Anita

MATCHING SIGNALS

name similarity
email / employee key
department + role
existing links

SOURCE B

person:A92
Anita Rao

DECISION	**MATCH**	merge to one canonical IRI; preserve both source IDs
	POSSIBLE	queue for human review; do not merge automatically
NO MATCH		keep separate entities

A false merge corrupts many edges; an unresolved duplicate fragments recall.

Source: Hogan et al. (2021)

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Mapping rules produce stable RDF triples

SOURCE RECORD

```
course_id      AML23702
title          Semantic
Web
faculty_id     F017
credits        3
```

MAPPING RULES

```
course IRI  = ex:course/{course_id}
faculty IRI = ex:person/{faculty_id}
title       → schema:name
credits     → ex:credits^^xsd:integer
```

GENERATED GRAPH

```
ex:course/AML23702  rdf:type      ex:Course .
ex:course/AML23702  schema:name   "Semantic Web" .
ex:person/F017       ex:teaches    ex:course/AML23702 .
ex:course/AML23702  ex:credits    "3"^^xsd:integer .
```

Prefer stable IRIs for entities; use typed literals for values; keep mappings versioned.

Sources: W3C RDF 1.1; R2RML; JSON-LD 1.1

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Validate, publish, observe, and refine the graph

1	→	2	→	3	→	4	→	5
VALIDATE		**LOAD**		**SERVE**		**OBSERVE**		**REFINE**
SHACL constraints reasoner consistency duplicate checks		RDF store / graph DB named graphs indexes		SPARQL / API search / applications access controls		quality metrics query failures source drift		mappings / schema identity rules new versions

MINIMUM RELEASE GATE

Valid schema • passing constraints • traceable provenance • acceptable identity precision • documented version

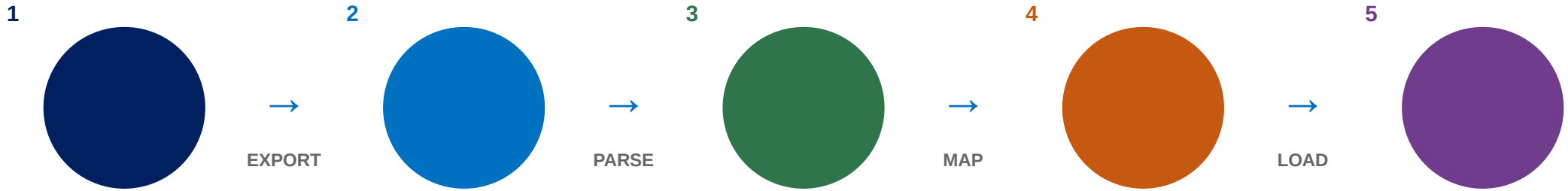
A knowledge graph is a maintained data product—not a one-time graph export.

Sources: W3C SHACL; SPARQL 1.1; PROV-O; Hogan et al. (2021)

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From Protégé ontology to a Neo4j knowledge graph

Automate parsing and mapping instead of hand-writing Cypher after every ontology change.



PROTÉGÉ

Model classes, properties and restrictions

OWL FILE

Export the ontology as university.owl

RDF/OWL PARSER

Read triples and ontology semantics

TRANSFORM

Map classes and properties to graph structures

NEO4J

Load a query-ready knowledge graph

WHAT THE MAPPING CREATES

Student — ENROLLED_IN → **Course** — OFFERED_BY → **Department**

Result: model once → map consistently → reload whenever the ontology changes.

Illustrative OWL-to-Neo4j transformation architecture based on the source content.